Software Requirements Specification for Flexin: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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15 Security Requirements

15.1 Access Requirements

ID	Requirement	Specification
SEC-ACC-01	Authentication and Sessions	**Description:** Users shall authenticate before accessing personal fitness data, portfolios, or transactions. The React Native client shall authenticate with the Spring API using secure token-based sessions. **Rationale:** Authentication protects personal data and virtual assets from unauthorized access. **Fit Criterion:** All protected API endpoints shall reject missing or invalid access tokens with HTTP 401. Access tokens shall expire within 60 minutes, with refresh or re-authentication required.
SEC-ACC-02	Role-Based Authorization	**Description:** Role-based access control shall restrict users to their own records and limit administrative or evaluator access to authorized functions. **Rationale:** Access restrictions prevent unauthorized modifications and cross-account data exposure. **Fit Criterion:** JUnit security tests shall confirm that attempts by standard users to access or modify another user's records return HTTP 403 in all tested cases.
SEC-ACC-03	Fitness Integration Authorization	**Description:** Flexin shall request only necessary read permissions for supported integrations, Android Health Connect and Garmin. Integration credentials shall be stored using secure, Android Keystore-backed storage. **Rationale:** Secure credential handling protects external accounts and limits access to fitness data. **Fit Criterion:** Integration tokens shall use Android Keystore-backed secure storage. No credentials, tokens, or secrets shall be committed to the project repository.

15.2 Integrity Requirements

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15.3 Privacy Requirements

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15.4 Audit Requirements

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15.5 Immunity Requirements

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17 Compliance Requirements

17.1 Legal Requirements

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17.2 Standards Compliance Requirements

ID	Requirement	Specification
COMP-STD-01	Android Platform Compliance	**Description:** The system shall comply with the Google Play Developer Program Policies. **Rationale:** The application is intended to be deployed on the Google Play Store as to be accessible on Android devices and as a result must conform to platform requirements. **Fit Criterion:** The application shall successfully pass all mandatory checks required for distribution through the Google Play Store.
COMP-STD-02	Accessibility Compliance	**Description:** The system shall comply with WCAG 2.1 Level AA accessibility guidelines for user-facing functionality. **Rationale:** The system should offer a simple and delightful user experience that should be accessible to users with varying abilities (i.e. user's with colour-blindness, hearing impairments, etc.). **Fit Criterion:** All major user workflows shall satisfy WCAG 2.1 Level AA success criteria as evaluated using automated accessibility tools and manual testing.
COMP-STD-03	TypeScript Coding Standards Compliance	**Description:** The frontend codebase shall comply with the Google TypeScript Style Guide. **Rationale:** Consistent coding standards improve maintainability, readability, and long-term project quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.

ID	Requirement	Specification
COMP-STD-04	Java Coding Standards Compliance	**Description:** The backend codebase shall comply with the Oracle Java Code Conventions. **Rationale:** Standardized coding practices improve maintainability, code review effectiveness, and software quality. **Fit Criterion:** All pull requests shall pass configured linting and formatting checks before merging into the main branch.
COMP-STD-05	Database Compliance	**Description:** The system shall store data in compliance with the MySQL Reference Manual and the SQL Standard (ISO/IEC 9075). **Rationale:** The project architecture relies upon MySQL as the primary relational database system. **Fit Criterion:** All schema definitions and queries will be tested to execute successfully on the supported MySQL version without compatibility warnings or errors.
COMP-STD-06	Software Testing Standard Compliance	**Description:** The system shall undergo thorough and complete testing in accordance with the IEEE 29119 Software Testing Standard. **Rationale:** Consistent testing practices reduce defects and improve software reliability. The system will comply with this standard to provide a unified, process-driven framework for software testing. **Fit Criterion:** All code merged into the main branch shall pass automated test execution in the CI/CD pipeline based on tests that were developed in accordance with the IEEE 29119 Software Testing Standard.

ID	Requirement	Specification
COMP-STD-07	Information Security Compliance	**Description:** The system shall be designed in accordance with the security principles outlined in ISO/IEC 27001. **Rationale:** The application stores personal, financial-learning, and fitness-related information that must be protected against unauthorized access and disclosure. **Fit Criterion:** Security controls for authentication, access control, data storage, and data transmission shall align with ISO/IEC 27001 recommendations.

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Insert your content here.

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26 Ideas for Solution

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27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?